

Volume 4: The Mathematics — Index

2026-05-19

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Target reader: Math referee, number theorist, Sarnak-class reader

Voice/style: Mathematical, proof-oriented

Chapters

Chapter	Topic	Contents
Ch01_Lie_Algebra_Verification.md	Lie Algebra Verification — The Isotropy Group $SO(5) \times SO(2)$	Originally LieAlgebraVerification.md (March 2026). Explicit numerical verification of $so(5,2)$ isotropy via 7×7 matrix representatives. Every structural claim confirmed by direct computation before abstract classification arguments invoked.
Ch02_Genesis_Bridge.md	Genesis and Bridge to Number Theory	Sections 27-31: Same Table, 40/40/20 Plan, Genesis (Light and Number), $Q^3 \subset Q^5$ and Riemann Hypothesis, From Winding to Zeta (Automorphic Structure)
Ch03_Why_This_Geometry.md	Why This Geometry (Part II)	Sections 32-37: Why Riemann, The 137/147 Pair, The Hunt, The Triple (D_{IV^5} Unique), Arithmetic Complexity ($P \neq NP$), Navier-Stokes
Ch04_Millennium_Problems.md	Millennium Problems and Unification	Sections 38-42: BSD, Hodge, Four-Color, Fermat and Poincaré, Unification (Silos Come Down)

Build instructions

Build the volume PDF with pandoc:

```
cd Working_Paper/Vol4_Mathematics
pandoc Ch*.md -o Volume.pdf --pdf-engine=xelatex -H ../notes/bst_pdf_header.tex
```

Navigation

- Up: Master Library Index

- All volumes: Vol 1 Journey, Vol 2 Framework, Vol 3 Physics, Vol 4 Mathematics, Vol 5 Predictions, Vol 6 Frontier

— Keeper, 2026-05-19